VIT Bhopal University

Student Performance Predictor using ML

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Name: Atharv Bisht

Registration No.: 25BCE10596

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1. Introduction

Student performance in examinations is influenced by several factors such as study habits, sleep patterns, attendance, and assignment performance. Analysing these factors can help in understanding how different behaviours affect academic results. With the help of Machine Learning, it is possible to identify patterns in student data and predict outcomes based on previous records.

The **Student Performance Predictor** is a machine learning based system designed to predict a student's exam score using factors such as study hours, sleep hours, attendance percentage, and assignment marks. The system stores student data in a dataset and uses it to train a machine learning model.

Objectives

The main objectives of the **Student Performance Predictor** project are:

- To develop a system that predicts student exam results using machine learning.
- To analyse how study habits and attendance affect academic performance.
- To gain practical experience with Python and machine learning libraries.
- To understand the workflow of building a machine learning application.

2. Tools and Technologies Used

The following tools and technologies were used in the project:

- **Python** – Used as the main programming language.
- **Pandas** – Used for handling and managing the dataset.
- **NumPy** – Used for numerical operations and data processing.
- **Scikit-learn** – Used to implement the machine learning model.
- **CSV File** – Used to store the dataset containing student records.
- **Visual Studio Code** – Used as the development environment.

3. Dataset Description

The dataset used in this project contains information related to student study habits and academic performance.

The dataset includes the following features:

- **Study Hours** – Number of hours a student studies per day.
- **Sleep Hours** – Average hours of sleep per day.
- **Attendance Percentage** – Percentage of classes attended.
- **Assignment Score** – Average marks obtained in assignments.
- **Exam Result** – Final exam marks obtained by the student.

The dataset grows as more student records are added to the system.

4. Machine Learning Model

The Student Performance Predictor uses the Random Forest Regressor algorithm from the Scikit-learn library.

Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions to improve accuracy. It is widely used for regression and prediction tasks.

The model is trained using the student dataset, and once trained, it can predict exam results based on the input features.

5. Implementation

The working process of the system is as follows:

1. The user runs the program using the main Python file.
2. A menu interface allows the user to select different options.
3. Users can add student records which are stored in the dataset.
4. The system trains the machine learning model using the stored data.
5. The user can enter new student details.
6. The trained model predicts the expected exam score.

6. Libraries Used

The **Student Performance Predictor** project uses several Python libraries to perform data handling, numerical computations, and machine learning tasks.

NumPy

NumPy is a numerical computing library used for handling arrays and performing mathematical operations efficiently. In this project, it is mainly used to convert input data into array format so that it can be processed by the machine learning model.

Pandas
Pandas is used for data manipulation and analysis. It allows the program to read and write CSV files, manage datasets, and perform operations such as loading student records from the dataset file.
Scikit-learn
Scikit-learn is a machine learning library that provides tools for building and training models. In this project, it is used to implement the Random Forest Regressor algorithm which is responsible for predicting the student's exam score.
CSV (Built-in Python Library)
The CSV module is used to store and manage student data in a CSV file format. It allows the program to easily save new student records and retrieve existing data for model training.

7. Module-wise Code Explanation

The Student Performance Predictor system is divided into multiple modules. Each module performs a specific task in the overall functioning of the program.

main.py

The main.py file acts as the central controller of the program. It manages the flow of the application and connects all other modules together.

When the program starts, the main.py file displays a menu interface to the user. The user can choose different options such as adding student data, viewing the dataset, training the machine learning model, or predicting a student's exam score.

Based on the user's choice, the program calls the appropriate module to perform the required operation. This file ensures that the entire system runs in a structured and organized way.

add_student.py

The add_student.py module is responsible for collecting student data from the user.

It asks the user to enter information such as:

- Study hours
- Sleep hours
- Attendance percentage
- Assignment marks
- Exam result

After receiving the inputs, the module stores the data in the dataset file (student_data.csv). This allows the dataset to grow over time as more student records are added.

data_loader.py

The data_loader.py module is used to load and display the dataset.

This module reads the CSV file containing student records using the Pandas library. It allows the program to access the stored data and display it to the user when required.

This module ensures that the dataset can be easily viewed and used for training the machine learning model.

train_model.py

The train_model.py module is responsible for training the machine learning model.

It loads the dataset from the CSV file and separates the input features from the target variable. The features include study hours, sleep hours, attendance, and assignment scores, while the target variable is the final exam result.

The module then trains a Random Forest Regressor model using this dataset. Once the training process is completed, the model is ready to make predictions.

predict_student.py

The predict_student.py module is used to predict the exam performance of a student.

This module takes input values from the user such as study hours, sleep hours, attendance percentage, and assignment scores. These inputs are converted into a format suitable for the machine learning model.

The trained model then analyzes the input data and predicts the expected exam score for the student.

student_data.csv

The student_data.csv file acts as the dataset for the system. It stores all student records that are entered through the program.

This dataset is used by the training module to train the machine learning model and improve prediction accuracy.

8. Results

1. After training the model, the system can predict exam scores based on new inputs.
2. Example Input:
3. Study Hours: 7
   Sleep Hours: 7
   Attendance: 85
   Assignment Score: 75
4. Predicted Output:
5. Predicted Exam Result: 79.92
6. This shows how the machine learning model estimates exam performance using the given features.

9. Limitations

Although the **Student Performance Predictor** demonstrates the application of machine learning in predicting student exam performance, the system has some limitations.

- The accuracy of predictions depends on the size and quality of the dataset used for training the model.
- The system currently uses only one machine learning algorithm, which may limit prediction performance.
- The program operates through a command-line interface and does not include a graphical user interface.
- Only a limited number of factors such as study hours, sleep hours, attendance, and assignment scores are considered, while many other factors may also influence student performance.

10. Future Scope

The **Student Performance Predictor** system can be improved in several ways in the future.

- Additional machine learning algorithms can be implemented to compare and improve prediction accuracy.
- A graphical user interface (GUI) can be developed to make the system more user-friendly.
- Larger and more realistic datasets can be used to improve the reliability of predictions.
- More features such as quiz scores, project marks, and class participation can be included to enhance the model.
- The system can be expanded to provide performance analysis and suggestions for improving student results

11. Conclusion

The **Student Performance Predictor** uses machine learning to estimate a student's exam score based on study hours, sleep hours, attendance, and assignment marks. The project demonstrates the basic workflow of data handling, model training, and prediction using Python.